SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PUBLIC DISTRIBUTION SYSTEM (PDS) USING AI AND IoT

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Abstract:

Food grain spoilage during storage is a major challenge in the Public Distribution System (PDS) due to environmental factors such as high temperature, humidity, excess moisture, and poor air quality, which reduce grain quality and lead to food wastage and economic loss. This project proposes a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System using Artificial Intelligence (AI) and the Internet of Things (IoT) to continuously monitor storage conditions and provide early warnings before spoilage occurs. The proposed system uses an ESP32 microcontroller integrated with a DHT22 temperature and humidity sensor, grain moisture sensor, and MQ135 gas sensor to collect real-time environmental data from grain storage facilities. The collected data is transmitted to an IoT cloud platform for continuous monitoring and analysis. A Machine Learning model processes the sensor data and predicts the spoilage risk by classifying the storage condition as Safe, Moderate Risk, or High Risk, enabling warehouse authorities to take timely preventive actions. The proposed solution aims to reduce food wastage, improve grain storage efficiency, minimize manual inspection, and support smart warehouse management in PDS godowns through a low-cost, scalable, and intelligent monitoring system.

Keywords: Artificial Intelligence, Internet of Things, Smart Grain Storage, Spoilage Risk Prediction, ESP32, Grain Moisture Monitoring, Public Distribution System

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